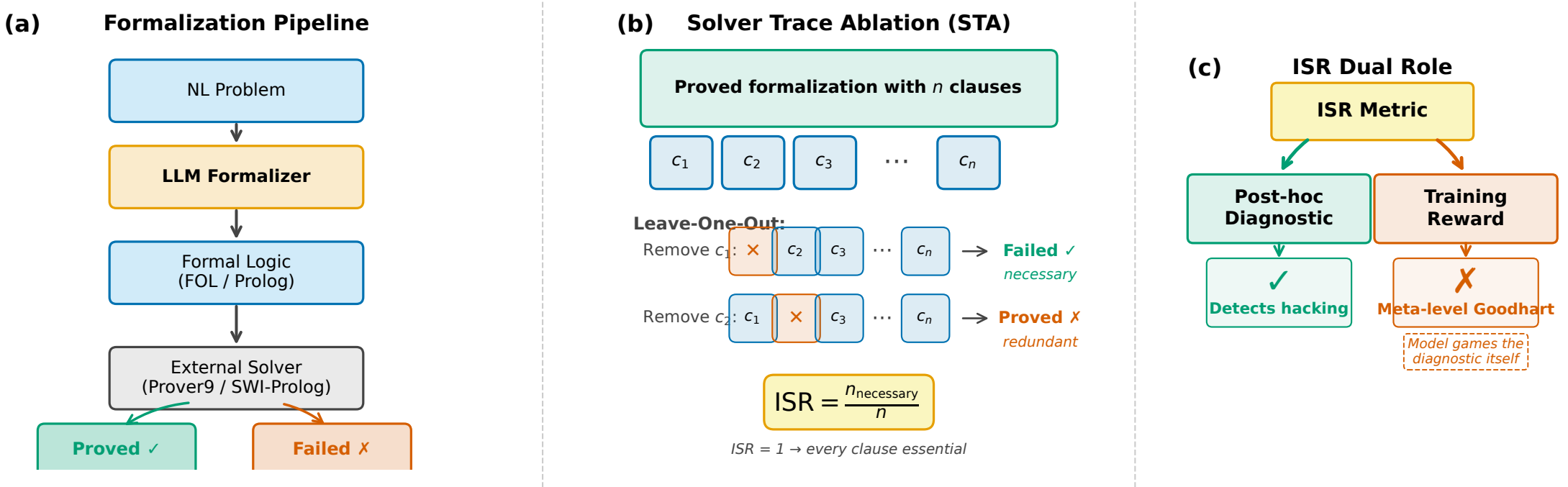
(a) Formalization Pipeline
NL Problem
LLM Formalizer
Formal Logic
(FOL / Prolog)
External Solver
(Prover9 / SWI-Prolog)
Proved ✓
Failed ✗
(b) Solver Trace Ablation (STA)
Proved formalization with n clauses
c1 c2 c3 ··· cn
Leave-One-Out:
Remove c1: × c2 c3 ··· cn → Failed ✓
necessary
Remove c2: c1 × c3 ··· cn → Proved ✗
redundant
ISR = n necessary / n
ISR = 1 → every clause essential
(c) ISR Dual Role
ISR Metric
Post-hoc Diagnostic
Training Reward
✓
Detects hacking
✗
Meta-level Goodhart
Model games the diagnostic itself